# Integración de funciones racionales de seno y coseno

Cuando el integrando es un cociente en términos de senos y/o cosenos, la sustitución $z = tan\left(\frac{x}{2}\right)$ lo convierte en una función racional de la variable $z$.

Sea

$$senx = 2sen\left(\frac{x}{2}\right)cos\left(\frac{x}{2}\right) = \frac{2sen\left(\frac{x}{2}\right)cos^2\left(\frac{x}{2}\right)}{cos\left(\frac{x}{2}\right)} = \frac{2sen\left(\frac{x}{2}\right)}{cos\left(\frac{x}{2}\right)} * \frac{1}{\frac{1}{cos^2\left(\frac{x}{2}\right)}} =$$

$$\frac{2tan\left(\frac{x}{2}\right)}{sec^2\left(\frac{x}{2}\right)} = \frac{2tan\left(\frac{x}{2}\right)}{1+tan^2\left(\frac{x}{2}\right)} = \frac{2z}{1+z^2}$$

$$cosx = 2cos^2\left(\frac{x}{2}\right) - 1 = \frac{2}{\frac{1}{cos^2\left(\frac{x}{2}\right)}} - 1 = \frac{2}{sec^2\left(\frac{x}{2}\right)} - 1$$

$$= \frac{2}{1 + tan^2\left(\frac{x}{2}\right)} - 1 = \frac{2 - 1 - tan^2\left(\frac{x}{2}\right)}{1 + tan^2\left(\frac{x}{2}\right)}$$

$$cosx = \frac{1-z^2}{1+z^2}$$

Y $dz = \frac{1}{2} sec^2\left(\frac{x}{2}\right) dx = \frac{1}{2}\left(1 + tan^2\left(\frac{x}{2}\right)\right) dx = \frac{1+z^2}{2} dx$

En resumen:

$$z = tan\left(\frac{x}{2}\right),\ senx = \frac{2z}{1+z^2}, cosx = \frac{1-z^2}{1+z^2}, dx = \frac{2}{1+z^2} dz$$

Veamos algunos ejemplos

1. Resolver $\int \frac{dx}{1+senx+cosx} = \int \frac{\frac{2}{1+z^2}}{1+\frac{2z}{1+z^2}+\frac{1-z^2}{1+z^2}} dz = \int \frac{\frac{2}{1+z^2}}{\frac{2z+2}{1+z^2}} dz =$

$$\int \frac{1}{z+1} dz = ln|z+1| + C$$

pero $z = tan\left(\frac{x}{2}\right)$

Así $\int \frac{dx}{1+senx+cosx} = ln\left|tan\left(\frac{x}{2}\right)+1\right| + C$

2. Resolver $\int \frac{dx}{4+2cos3x}$ Sea $u = 3x, du = 3dx$ entonces

$$\int \frac{dx}{4+2cos3x}$$

$$= \frac{1}{3}\int \frac{du}{4+2cosu} = \frac{1}{6}\int \frac{du}{2+cosu} = \frac{1}{6}\int \frac{\frac{2}{1+z^2}}{2-\frac{1-z^2}{1+z^2}}dz$$

$$= \frac{1}{6}\int \frac{\frac{2}{1+z^2}}{\frac{3+z^2}{1+z^2}}dz$$

$$= \frac{1}{6}\int \frac{2}{3+z^2}dz = \frac{1}{3}\frac{1}{\sqrt{3}}tan^{-1}\left(\frac{z}{\sqrt{3}}\right) + C$$

$$= \frac{\sqrt{3}}{9}tan^{-1}\left(\frac{1}{\sqrt{3}}tan\left(\frac{u}{2}\right)\right) + C$$

Así $\int \frac{dx}{4+2cos3x} = \frac{\sqrt{3}}{9}tan^{-1}\left(\frac{1}{\sqrt{3}}tan\left(\frac{3x}{2}\right)\right) + C$

3. Resolver $\int \frac{1}{tanx+senx}dx = \int \frac{1}{\frac{senx}{cosx}+senx}dx =$

$$\int \frac{cosx}{senx+senxcosx}dx = \int \frac{cosx}{senx(1+cosx)}dx = \int \frac{\frac{1-z^2}{1+z^2}*\frac{2}{1+z^2}}{\frac{2z}{1+z^2}\left(1+\frac{1-z^2}{1+z^2}\right)}dz =$$

$$\int \frac{\frac{2(1-z^2)}{(1+z^2)^2}}{\frac{2z}{(1+z^2)^2}(1+z^2+1-z^2)}dz = \frac{1}{2}\int \frac{(1-z^2)}{z}dz = \frac{1}{2}\left(ln|z| - \frac{z^2}{2}\right) + C$$

Pero $z = tan\left(\frac{x}{2}\right)$

Así $\int \frac{1}{tanx+senx}dx = \frac{1}{2}\left(ln\left|tan\left(\frac{x}{2}\right)\right| - \frac{\left(tan\left(\frac{x}{2}\right)\right)^2}{2}\right) + C$

Ejercicios propuestos

$$\int \frac{senx}{senx-tanx}\,dx \qquad \int \frac{1}{senx+cosx}\,dx \qquad \int \frac{senx-cosx}{senx+cosx}\,dx$$

$$\int \frac{1}{cot2x(1-cos2x)}\,dx \qquad \int \frac{1}{5senx+4sen^2x}\,dx$$